

Load Interconnection Request System Impact Study Report

For Queue Project Q22-2254

Woodland Tap 345 kV
August 28, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2254. The Interconnection Customer proposes a 593.3 MW large load (data center) facility interconnecting at the Woodland Tap 345 kV substation in the Pennsylvania area. The request was received on December 15, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 535 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$13,721,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Youngwood 500/345 kV Transformer Bank	Youngwood - Univ Of Nc Chapel Hill Cogen Facility 345kV Line	101.6	873	22.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Youngwood 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$23,990,000	57% (\$13,721,000)
	Total allocated to project		\$13,721,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.